

MARDIE, AustraliaWMO: **943060**Lat: **21.191S**Lon: **115.980E**Elev: **12**StdP: **101.18**Time Zone: **8.00 (E08)**Period: **00-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	11.6	-8.6	1.8	26.7	-6.0	2.3	26.0	11.0	23.6	10.1	24.3	3.3	120	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	9.6	41.0	21.4	39.3	21.8	38.0	22.0	27.8	30.2	27.2	30.5	26.6	30.6	5.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.4	23.3	28.4	26.6	22.1	28.3	25.7	21.0	28.5	89.2	30.1	86.2	30.3	83.8	30.6	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.5	8.7	DB	8.1	44.8	0.9	2.0	7.4	46.2	6.8	47.4	6.3	48.5	5.6	50.0	(4)
(5)			WB	3.9	28.7	1.4	1.1	2.9	29.5	2.0	30.1	1.2	30.7	0.2	31.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.2	30.8	30.9	30.9	28.6	24.2	21.0	20.2	20.9	23.1	26.6	28.0	30.0	(6)	
	DBStd	4.62	2.51	2.23	2.13	2.15	2.34	2.31	2.04	1.76	2.02	2.75	2.57	2.63	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	13	0	0	0	0	0	4	7	2	0	0	0	0	(9)	
	CDD10.0	5923	644	586	647	558	439	328	317	339	392	515	539	619	(10)	
	CDD18.3	2895	386	352	389	308	181	83	66	82	142	257	289	361	(11)	
	CDH23.3	35507	5208	4822	5317	3701	1703	678	613	896	1506	2976	3385	4702	(12)	
(13)	CDH26.7	18495	2886	2702	3017	1979	720	159	134	256	646	1604	1793	2601	(13)	
(14)	Wind	WSAvg	4.2	4.7	4.4	4.0	3.5	3.5	3.8	4.0	4.2	4.5	5.0	4.8	(14)	
(15) Precipitation	PrecAvg	322	41	81	50	18	42	47	13	7	1	1	1	12	(15)	
	PrecMax	857	241	675	299	146	212	236	91	55	10	12	10	171	(16)	
	PrecMin	84	0	0	0	0	0	0	0	0	0	0	0	0	(17)	
	PrecStd	174	61	112	74	33	54	59	18	13	3	2	2	31	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.7	42.3	41.7	39.5	35.7	31.2	31.1	32.3	36.3	39.9	41.5	42.5	(19)	
		MCWB	20.9	22.9	21.7	20.5	17.9	17.1	15.8	14.8	17.3	18.4	19.2	20.8	(20)	
	2%	DB	40.0	39.6	39.5	37.7	33.9	29.9	29.5	30.6	34.0	37.8	38.7	39.8	(21)	
		MCWB	23.1	23.2	22.5	20.0	17.6	16.2	15.5	15.1	16.9	18.5	20.0	21.3	(22)	
	5%	DB	37.6	37.6	37.7	36.3	32.3	28.6	28.2	29.3	32.2	36.1	36.5	37.6	(23)	
		MCWB	24.0	24.0	23.0	20.1	17.6	15.9	15.1	15.0	16.7	19.1	20.2	21.9	(24)	
	10%	DB	35.8	35.9	36.2	34.8	30.7	27.3	26.8	28.1	30.4	34.2	34.5	35.8	(25)	
		MCWB	24.4	24.6	23.2	20.2	17.5	15.7	15.0	14.9	16.6	18.7	20.4	22.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	28.6	28.8	28.2	27.7	25.4	24.2	21.5	21.3	21.9	23.8	25.3	27.7	(27)	
		MCDB	30.5	30.4	29.5	28.8	26.3	24.9	24.5	24.4	25.8	28.3	28.7	29.7	(28)	
	2%	WB	27.6	27.8	27.3	25.9	23.5	22.2	19.7	19.6	20.5	22.6	24.5	26.6	(29)	
		MCDB	30.3	31.1	30.0	29.1	26.2	24.3	23.0	24.1	26.0	28.6	29.2	31.1	(30)	
	5%	WB	27.0	27.1	26.7	24.9	22.2	20.6	18.5	18.6	19.7	21.8	23.7	25.7	(31)	
		MCDB	30.5	31.0	30.3	29.0	25.8	23.7	23.3	23.9	25.8	28.7	29.2	30.7	(32)	
	10%	WB	26.4	26.5	26.1	24.1	21.2	19.3	17.4	17.5	18.9	21.1	22.9	25.0	(33)	
		MCDB	30.7	31.0	30.5	28.7	25.5	23.2	23.2	23.7	25.6	28.8	29.3	30.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.5	9.6	10.5	12.5	13.0	12.7	13.9	15.0	15.2	14.7	12.7	11.0	(35)	
		MCDBR	13.0	12.7	13.2	14.5	15.0	14.0	15.6	16.6	17.7	17.1	15.8	14.5	(36)	
	5% WB	MCWBR	5.1	4.7	5.2	5.9	5.5	5.6	6.2	6.5	7.0	6.8	6.4	5.8	(37)	
		MCWBR	9.1	9.9	10.3	11.0	10.5	8.8	11.7	13.2	14.7	14.3	12.6	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.400	0.396	0.378	0.354	0.333	0.322	0.314	0.317	0.333	0.352	0.357	0.374	(40)	
		taud	2.497	2.525	2.562	2.588	2.582	2.594	2.595	2.544	2.499	2.471	2.494	2.520	(41)	
	Ebn at Noon	Ebn at Noon	945	937	929	912	895	886	906	936	958	967	980	971	(42)	
		Edh at Noon	116	111	103	93	86	82	84	95	107	116	116	114	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	7.57	7.20	6.40	5.50	4.62	4.17	4.52	5.61	6.68	7.55	8.09	8.12	(44)	
	RadStd	RadStd	0.42	0.47	0.43	0.31	0.36	0.30	0.31	0.17	0.24	0.20	0.22	0.34	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page